P-02: PAH binding and reporter model results

Task: Calculate binding, reporter and competition outputs from the source model parameters.

Inputs: Parameter JSON and three documentary source model scripts.

Motivation: Explain the behaviour of the existing equations.

Goal: Export binding curves, reporter accumulation and mixture occupancy.

Prediction: Half occupancy occurs at free ligand = Kd; fixed competitors shift the pyrene occupancy curve to higher concentrations.

```
from pathlib import Path
import hashlib, json, platform, sys
from importlib.metadata import version
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

from scipy.integrate import solve_ivp
TASK_ID = "P-02"

PROJECT_ROOT = next((p for p in [Path.cwd().resolve(), *Path.cwd().resolve().parents]
                     if (p / "data/input").is_dir() and (p / "tasks").is_dir()), None)
if PROJECT_ROOT is None:
    raise FileNotFoundError("Start Jupyter inside the project folder.")
INPUT = PROJECT_ROOT / "data/input"
OUTPUT = PROJECT_ROOT / "data/output" / TASK_ID
OUTPUT.mkdir(parents=True, exist_ok=True)
BLUE, ORANGE = "#0072B2", "#E69F00"
plt.rcParams.update({"font.family": "sans-serif", "font.size": 9,
                     "axes.spines.top": False, "axes.spines.right": False,
                     "savefig.dpi": 300})

ASSUMPTIONS_FILE = "P02_source_assumptions.json"
SOURCE_FILES = ["P02_model1_binding.py", "P02_model2_reporter.py", "P02_model3_competition.py"]
ALL_INPUTS = [ASSUMPTIONS_FILE] + SOURCE_FILES
FREE_LIGAND_NM = np.logspace(-2, 6, 240)
REPORTER_TIMES_MIN = np.linspace(0, 1440, 1441)
INITIAL_STATE = [0.01, 0.0, 0.0, 0.0]
```

```
COMMIT = "3539358236841e3f95f3e9efdc6d954fe1a1eca2"
SOURCE_URL = f"https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/{COMMIT}/tools/autodock-vina/md/models"
```

Materials and methods: binding

For one site, occupancy is θ=L/(Kd+L). Set koff=kon·Kd. The time to 95% approach from zero occupancy is -ln(0.05)/[kon·(L+Kd)]. The source parameter file defines the scenarios.

Scope and working assumptions. Affinities, concentrations and rates are fixed source-assigned parameters. The calculations assume one site, fixed free ligand, mutually exclusive competition and no depletion. The reporter represents an E. coli CFP/YFP model. Results are numerical consequences of these equations and parameters, rather than fitted measurements of yeast or cadmium performance.

```
assumptions = json.loads((INPUT / ASSUMPTIONS_FILE).read_text(encoding="utf-8"))
source_hashes = {name: hashlib.sha256((INPUT / name).read_bytes()).hexdigest() for name in ALL_INPUTS}
binding_rows, kinetic_rows = [], []
kon = assumptions["model1"]["kon_nM_inv_s_inv"]
for kd in assumptions["model1"]["Kd_grid_nM"]:
    occupancy = FREE_LIGAND_NM / (kd + FREE_LIGAND_NM)
    binding_rows.extend({"free_ligand_nM": l, "assumed_Kd_nM": kd, "occupancy": th} for l, th in zip(FREE_LIGAND_NM, occupancy))
    for ligand_nm in [1.0, 100.0]:
        kinetic_rows.append({"assumed_Kd_nM": kd, "free_ligand_nM": ligand_nm,
            "t95_min": -np.log(0.05) / (kon * (ligand_nm + kd)) / 60})
binding, kinetics = pd.DataFrame(binding_rows), pd.DataFrame(kinetic_rows)
binding.to_csv(OUTPUT / "binding_curve_plot_data.csv", index=False)
kinetics.to_csv(OUTPUT / "binding_kinetics.csv", index=False)
# A long table keeps every assigned parameter visible in the PDF.
parameter_table = pd.json_normalize(assumptions, sep=".").T.reset_index()
parameter_table.columns = ["Parameter", "Assumed value"]
parameter_table["Input data"] = ASSUMPTIONS_FILE
parameter_table["Finished without more wet-lab data?"] = "Yes: conditional model"
display(parameter_table)
parameter_table.to_csv(OUTPUT / "assumed_parameters.csv", index=False)
```

	Parameter	Assumed value	Input data	Finished w ithout more w et-lab data?
0	model1.Kd_grid_nM	[1.0, 10.0, 100.0, 1000.0, 10000.0]	P02_source_assumptions.json	Yes: conditional model
1	model1.kon_nM_inv_s_inv	0.00001	P02_source_assumptions.json	Yes: conditional model
2	model2.Kd_nM.fluoranthene	100.0	P02_source_assumptions.json	Yes: conditional model
3	model2.Kd_nM.pyrene	300.0	P02_source_assumptions.json	Yes: conditional model
4	model2.Kd_nM.phenanthrene	500.0	P02_source_assumptions.json	Yes: conditional model
5	model2.Kd_nM.naphthalene	5000.0	P02_source_assumptions.json	Yes: conditional model
6	model2.R_free	1.0	P02_source_assumptions.json	Yes: conditional model
7	model2.R_bound	1.6	P02_source_assumptions.json	Yes: conditional model
8	model2.parameters.alpha_arbitrary_per_min	1.0	P02_source_assumptions.json	Yes: conditional model
9	model2.parameters.mRNA_half_life_min	5	P02_source_assumptions.json	Yes: conditional model
10	model2.parameters.translation_per_mRNA_per_min	2.0	P02_source_assumptions.json	Yes: conditional model
11	model2.parameters.maturation_half_life_min	30	P02_source_assumptions.json	Yes: conditional model
12	model2.parameters.doubling_time_min	40	P02_source_assumptions.json	Yes: conditional model
13	model2.parameters.turnover_half_life_min	1440	P02_source_assumptions.json	Yes: conditional model
14	model2.parameters.stationary_promoter_floor	0.02	P02_source_assumptions.json	Yes: conditional model
15	model2.parameters.kon_nM_inv_min_inv	0.0006	P02_source_assumptions.json	Yes: conditional model
16	model3.assumed_Kd_nM.fluoranthene	100.0	P02_source_assumptions.json	Yes: conditional model
17	model3.assumed_Kd_nM.pyrene	300.0	P02_source_assumptions.json	Yes: conditional model
18	model3.assumed_Kd_nM.phenanthrene	500.0	P02_source_assumptions.json	Yes: conditional model
19	model3.assumed_Kd_nM.anthracene	1500.0	P02_source_assumptions.json	Yes: conditional model
20	model3.assumed_Kd_nM.naphthalene	5000.0	P02_source_assumptions.json	Yes: conditional model
21	model3.assumed_Kd_nM.glucose	1000000.0	P02_source_assumptions.json	Yes: conditional model
22	model3.assumed_free_concentration_nM.pyrene	20.0	P02_source_assumptions.json	Yes: conditional model
23	model3.assumed_free_concentration_nM.phenanthrene	50.0	P02_source_assumptions.json	Yes: conditional model
24	model3.assumed_free_concentration_nM.fluoranthene	20.0	P02_source_assumptions.json	Yes: conditional model
25	model3.assumed_free_concentration_nM.anthracene	10.0	P02_source_assumptions.json	Yes: conditional model
26	model3.assumed_free_concentration_nM.naphthalene	500.0	P02_source_assumptions.json	Yes: conditional model
27	model3.assumed_free_concentration_nM.glucose	1000.0	P02_source_assumptions.json	Yes: conditional model

Materials and methods: reporter accumulation

N is growth relative to carrying capacity; m, I and S are mRNA, immature and mature reporter in model units.

$\mu=\mu_0(1-N)$; $a=\alpha[\text{basal}+(1-\text{basal})\mu/\mu_0]$. The ODEs are $N'=\mu N$; $m'=a-\delta_m m$; $I'=\beta m-(k_{mat}+\mu+\delta_p)I$; $S'=k_{mat}\cdot I-(\mu+\delta_p)S$. Half-times define the decay rates. Integrate 0–1440 min with DOP853, relative tolerance 10^{-8} and absolute tolerance 10^{-10}. Report the maximum on the sampled one-minute grid. The ideal ratio is $R_{free}+(R_{bound}-R_{free})L/(K_d+L)$.

```
model2, p = assumptions["model2"], assumptions["model2"]["parameters"]
delta_m = np.log(2) / p["mRNA_half_life_min"]
kmat = np.log(2) / p["maturation_half_life_min"]
mu0 = np.log(2) / p["doubling_time_min"]
delta_p = np.log(2) / p["turnover_half_life_min"]
def reporter_rhs(time, state):
    # Growth-dependent production, maturation, dilution and turnover.
    n, mrna, immature, mature = state
    mu = mu0 * (1 - n)
    production = p["alpha_arbitrary_per_min"] * (p["stationary_promoter_floor"] + (1 - p["stationary_promoter_floor"]) * mu / mu0)
    return [mu * n, production - delta_m * mrna,
            p["translation_per_mRNA_per_min"] * mrna - (kmat + mu + delta_p) * immature,
            kmat * immature - (mu + delta_p) * mature]
solution = solve_ivp(reporter_rhs, (0, 1440), INITIAL_STATE, t_eval=REPORTER_TIMES_MIN,
                     method="DOP853", rtol=1e-8, atol=1e-10)
if not solution.success:
    raise RuntimeError(solution.message)
reporter = pd.DataFrame(solution.y.T, columns=["growth_fraction", "mrna_model_units", "immature_model_units", "mature_model_units"])
reporter.insert(0, "time_min", solution.t)
peak_index = reporter.mature_model_units.idxmax()
peak_time_h = reporter.loc[peak_index, "time_min"] / 60
reporter["mature_fraction_of_model_max"] = reporter.mature_model_units / reporter.mature_model_units.max()
ratio_rows = []
for ligand, kd in model2["Kd_nM"].items():
    for concentration in [20.0, 100.0, 1000.0]:
        ratio_rows.append({"ligand": ligand, "free_ligand_nM": concentration, "assumed_Kd_nM": kd,
            "ideal_ratio": model2["R_free"] + (model2["R_bound"] - model2["R_free"]) * concentration / (kd + concentration)})
ratios = pd.DataFrame(ratio_rows)
reporter.to_csv(OUTPUT / "reporter_plot_data.csv", index=False)
ratios.to_csv(OUTPUT / "ideal_reporter_ratios.csv", index=False)
```

Materials and methods: competition

Ligands compete for one mutually exclusive site at fixed free concentrations. For ligand i, $\theta_i=(L_i/K_{d,i})/(1+\Sigma L_j/K_{d,j})$; the unbound fraction is $1/(1+\Sigma L_j/K_{d,j})$. The apparent pyrene half-site concentration under fixed competitors is $K_{d,\text{pyrene}}\cdot(1+\Sigma_{\text{competitors}} L_j/K_{d,j})$.

```
model3 = assumptions["model3"]
kd = pd.Series(model3["assumed_Kd_nM"], dtype=float)
concentration = pd.Series(model3["assumed_free_concentration_nM"], dtype=float).reindex(kd.index)
assert (kd > 0).all() and (concentration >= 0).all()
weight = concentration / kd
mixture = pd.DataFrame({"assumed_Kd_nM": kd, "assumed_free_ligand_nM": concentration,
                        "competition_weight": weight, "site_fraction": weight / (1 + weight.sum())})
mixture.index.name = "ligand"
unbound = 1 / (1 + weight.sum())
competitor_weight = weight.drop("pyrene").sum()
apparent_half_nM = kd["pyrene"] * (1 + competitor_weight)
competition = pd.DataFrame({"free_pyrene_nM": FREE_LIGAND_NM,
    "pyrene_alone": FREE_LIGAND_NM / (kd["pyrene"] + FREE_LIGAND_NM),
    "fixed_competitors": FREE_LIGAND_NM / (apparent_half_nM + FREE_LIGAND_NM)})
mixture.to_csv(OUTPUT / "mixture_plot_data.csv")
competition.to_csv(OUTPUT / "pyrene_competition_curves.csv", index=False)
```

Results

The figures and CSV files give the outputs of each model. Reporter accumulation is divided by its sampled maximum for plotting.

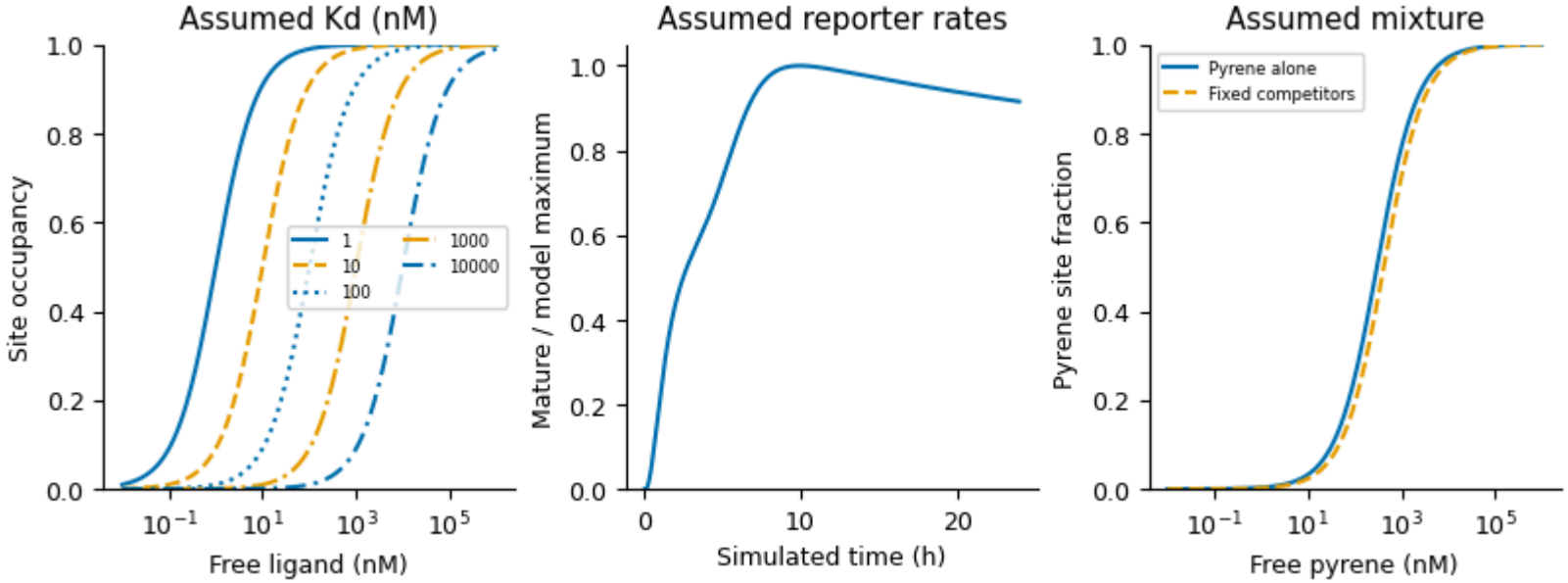
```
def result_view(table, input_data):
    return table.assign(**{"Input data": input_data, "Finished without more wet-lab data?": "Yes: conditional model"})
display(result_view(kinetics.round(4), "Model 1 assumptions"),
        result_view(ratios.round(4), "Model 2 assumptions"),
        result_view(mixture.round(5), "Model 3 assumptions"))
fig, axes = plt.subplots(1, 3, figsize=(8.0, 3.0), layout="constrained")
for index, (kd_value, group) in enumerate(binding.groupby("assumed_Kd_nM")):
    axes[0].semilogx(group.free_ligand_nM, group.occupancy, label=f"{kd_value:g}", color=[BLUE, ORANGE][index % 2], linestyle=["-", "--", ":", "-.", (0, (5, 2, 1,
2))][index])
axes[0].set(xlabel="Free ligand (nM)", ylabel="Site occupancy", ylim=(0, 1), title="Assumed Kd (nM)")
axes[0].legend(fontsize=6, ncol=2)
axes[1].plot(reporter.time_min / 60, reporter.mature_fraction_of_model_max, color=BLUE)
axes[1].set(xlabel="Simulated time (h)", ylabel="Mature / model maximum", ylim=(0, 1.05), title="Assumed reporter rates")
axes[2].semilogx(competition.free_pyrene_nM, competition.pyrene_alone, color=BLUE, label="Pyrene alone")
axes[2].semilogx(competition.free_pyrene_nM, competition.fixed_competitors, color=ORANGE, ls="--", label="Fixed competitors")
axes[2].set(xlabel="Free pyrene (nM)", ylabel="Pyrene site fraction", ylim=(0, 1), title="Assumed mixture")
axes[2].legend(fontsize=6)
fig.savefig(OUTPUT / "conditional_models.png", dpi=300, bbox_inches="tight")
fig.savefig(OUTPUT / "conditional_models.svg", bbox_inches="tight")
plt.show()
runtime = {name: version(name) for name in ["numpy", "pandas", "scipy", "matplotlib"]}
(OUTPUT / "source_manifest.json").write_text(json.dumps({"sha256": source_hashes,
```

```
    "source_commit": COMMIT, "python": sys.version, "platform": platform.platform(),
    "packages": runtime, "assumptions": assumptions}, indent=2), encoding="utf-8");
```

	assumed_Kd_nM	free_ligand_nM	t95_min	Input data	Finished w ithout more w et-lab data?
0	1.0	1.0	2496.4436	Model 1 assumptions	Yes: conditional model
1	1.0	100.0	49.4345	Model 1 assumptions	Yes: conditional model
2	10.0	1.0	453.8988	Model 1 assumptions	Yes: conditional model
3	10.0	100.0	45.3899	Model 1 assumptions	Yes: conditional model
4	100.0	1.0	49.4345	Model 1 assumptions	Yes: conditional model
5	100.0	100.0	24.9644	Model 1 assumptions	Yes: conditional model
6	1000.0	1.0	4.9879	Model 1 assumptions	Yes: conditional model
7	1000.0	100.0	4.5390	Model 1 assumptions	Yes: conditional model
8	10000.0	1.0	0.4992	Model 1 assumptions	Yes: conditional model
9	10000.0	100.0	0.4943	Model 1 assumptions	Yes: conditional model

	ligand	free_ligand_nM	assumed_Kd_nM	ideal_ratio	Input data	Finished w ithout more w et-lab data?
0	fluoranthene	20.0	100.0	1.1000	Model 2 assumptions	Yes: conditional model
1	fluoranthene	100.0	100.0	1.3000	Model 2 assumptions	Yes: conditional model
2	fluoranthene	1000.0	100.0	1.5455	Model 2 assumptions	Yes: conditional model
3	pyrene	20.0	300.0	1.0375	Model 2 assumptions	Yes: conditional model
4	pyrene	100.0	300.0	1.1500	Model 2 assumptions	Yes: conditional model
5	pyrene	1000.0	300.0	1.4615	Model 2 assumptions	Yes: conditional model
6	phenanthrene	20.0	500.0	1.0231	Model 2 assumptions	Yes: conditional model
7	phenanthrene	100.0	500.0	1.1000	Model 2 assumptions	Yes: conditional model
8	phenanthrene	1000.0	500.0	1.4000	Model 2 assumptions	Yes: conditional model
9	naphthalene	20.0	5000.0	1.0024	Model 2 assumptions	Yes: conditional model
10	naphthalene	100.0	5000.0	1.0118	Model 2 assumptions	Yes: conditional model
11	naphthalene	1000.0	5000.0	1.1000	Model 2 assumptions	Yes: conditional model

ligand	assumed_Kd_nM	assumed_free_ligand_nM	competition_weight	site_fraction	Input data	Finished without more wet-lab data?
fluoranthene	100.0	20.0	0.20000	0.13565	Model 3 assumptions	Yes: conditional model
pyrene	300.0	20.0	0.06667	0.04522	Model 3 assumptions	Yes: conditional model
phenanthrene	500.0	50.0	0.10000	0.06783	Model 3 assumptions	Yes: conditional model
anthracene	1500.0	10.0	0.00667	0.00452	Model 3 assumptions	Yes: conditional model
naphthalene	5000.0	500.0	0.10000	0.06783	Model 3 assumptions	Yes: conditional model
glucose	1000000.0	1000.0	0.00100	0.00068	Model 3 assumptions	Yes: conditional model




```
# The conclusion stays conditional on the explicitly assigned inputs.
t95 = float(kinetics[(kinetics.assumed_Kd_nM == 100) & (kinetics.free_ligand_nM == 100)].t95_min.iloc[0])
pyrene_ratio = float(ratios[(ratios.ligand == "pyrene") & (ratios.free_ligand_nM == 100)].ideal_ratio.iloc[0])
pyrene_fraction = float(mixture.loc["pyrene", "site_fraction"])
conclusion = ((f"At Kd = free ligand = 100 nM, the binding model gives t95 = {t95:.3f} min. "
               f"The reporter reaches its sampled maximum at {peak_time_h:.3f} simulated hours. "
               f"Fixed competitors shift pyrene half occupancy from {kd['pyrene']:.1f} to {apparent_half_nM:.1f} nM; "
               f"mixture pyrene occupancy is {pyrene_fraction:.5f} and the unbound fraction is {unbound:.5f}."))
table_rows = [["Binding: Kd=L=100 nM", "t95 (min)", round(t95, 4)], ["Reporter", "Sampled peak (simulated h)", round(peak_time_h, 4)],
              ["Pyrene at 100 nM", "Ideal ratio", round(pyrene_ratio, 4)], ["Assumed mixture", "Pyrene site fraction", round(pyrene_fraction, 6)],
              ["Assumed mixture", "Unbound fraction", round(float(unbound), 6)], ["Fixed competitors", "Pyrene half-site (nM)", round(float(apparent_half_nM), 4)]]
```

```
table_rows = [row + ["Stored assumed parameters", "Yes: conditional model"] for row in table_rows]
table_columns = ["Scenario", "Quantity", "Value", "Input data", "Finished without more wet-lab data?"]
display(pd.DataFrame(table_rows, columns=table_columns))
pd.DataFrame(table_rows, columns=table_columns).to_csv(OUTPUT / "scenario_summary.csv", index=False)
summary = {"title": "P-02: PAH binding and reporter model results", "question": "What follows from the project's existing assumed model inputs?",
    "materials": ["Stored source_assumptions.json; original model scripts preserved as inert equation/provenance records.", "No fitted parameters or new experimental data."],
    "methods": ["Analytical single-site occupancy and fixed-ligand approach to equilibrium.", "Source reporter ODE integrated by DOP853 (rtol 1e-8; atol 1e-10) on a one-minute output grid.", "Deterministic competitive occupancy at assigned free concentrations; no score-to-Kd conversion."],
    "results": [f"Model 1, Kd=L=100 nM: t95={t95:.4f} min.", f"Reporter sampled maximum: {peak_time_h:.4f} simulated h; ideal pyrene ratio at 100 nM={pyrene_ratio:.4f}.", f"Model 3 mixture: pyrene fraction={pyrene_fraction:.6f}, unbound={unbound:.6f}; competitor half-site={apparent_half_nM:.1f} nM."],
    "conclusion": conclusion,
    "limitations": [],
    "figures": [{"file": "conditional_models.png", "caption": "Conditional binding curves, reporter accumulation and pyrene competition under stored assumptions. All times are simulated; lines are not measured data."}],
    "references": [SOURCE_URL, "Stored source_assumptions.json: explicit source-derived parameter transcription; source hashes and complete parameters are in source_manifest.json."],
    "table": {"columns": table_columns, "rows": table_rows}}
summary["scope_and_assumptions"] = 'Affinities, concentrations and rates are fixed source-assigned parameters. The calculations assume one site, fixed free ligand, mutually exclusive competition and no depletion. The reporter represents an E. coli CFP/YFP model. Results are numerical consequences of these equations and parameters, rather than fitted measurements of yeast or cadmium performance.'
summary.pop("limitations", None)
(OUTPUT / "summary.json").write_text(json.dumps(summary, indent=2, ensure_ascii=False), encoding="utf-8")
display(Markdown("## Conclusion and interpretation\n\n" + conclusion))
```

	Scenario	Quantity	Value	Input data	Finished w ithout more w et-lab data?
0	Binding: Kd=L=100 nM	t95 (min)	24.964400	Stored assumed parameters	Yes: conditional model
1	Reporter	Sampled peak (simulated h)	9.933300	Stored assumed parameters	Yes: conditional model
2	Pyrene at 100 nM	Ideal ratio	1.150000	Stored assumed parameters	Yes: conditional model
3	Assumed mixture	Pyrene site fraction	0.045218	Stored assumed parameters	Yes: conditional model
4	Assumed mixture	Unbound fraction	0.678273	Stored assumed parameters	Yes: conditional model
5	Fixed competitors	Pyrene half-site (nM)	422.300000	Stored assumed parameters	Yes: conditional model

Conclusion and interpretation

At Kd = free ligand = 100 nM, the binding model gives t95 = 24.964 min. The reporter reaches its sampled maximum at 9.933 simulated hours. Fixed competitors shift pyrene half occupancy from 300.0 to 422.3 nM; mixture pyrene occupancy is 0.04522 and the unbound fraction is 0.67827.